CS & IT ENGINEERING

DIGITAL LOGIC

Combinational Circuit



Lecture No. 08



By- CHANDAN SIR



01 HA

02 FA

03 Question Practice

04 Discussion

Half Adder

→ Two bit adder

0	0	1	1
+0	+1	+0	+1
00	01	01	10

carry, sum, carry, sum

	1
	+1
1	0

Two bit adder are known as half adder.

Half Adder

Step 1. INPUTS & OUTPUTS



Half Adder

Step 2. TRUTH TABLE

A	B	Sum	Carry
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Half Adder

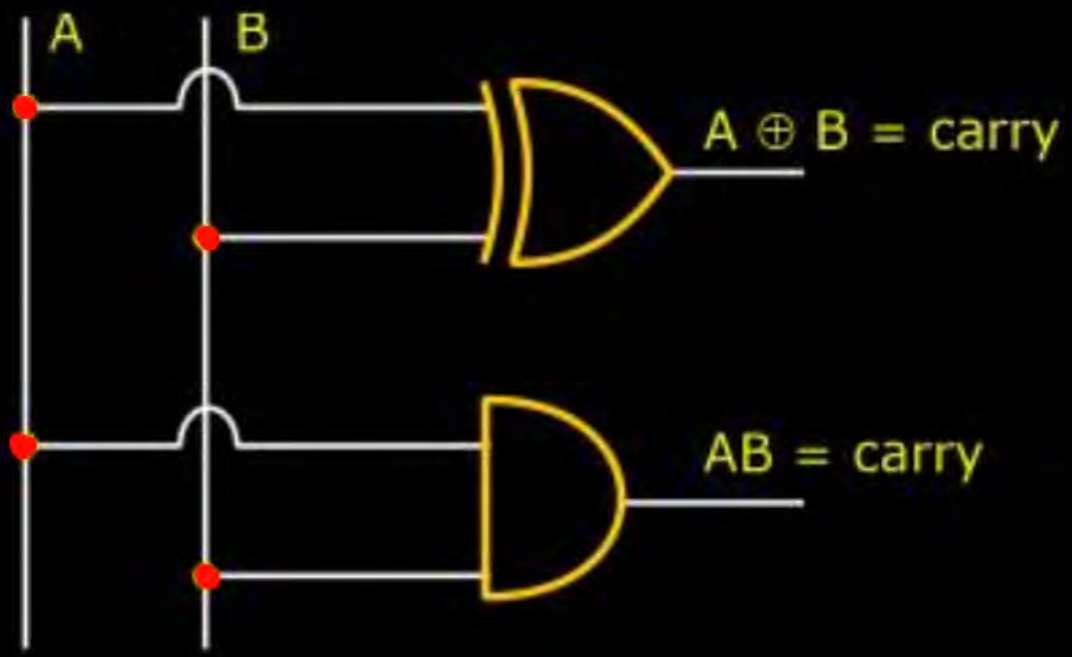
Step 3.

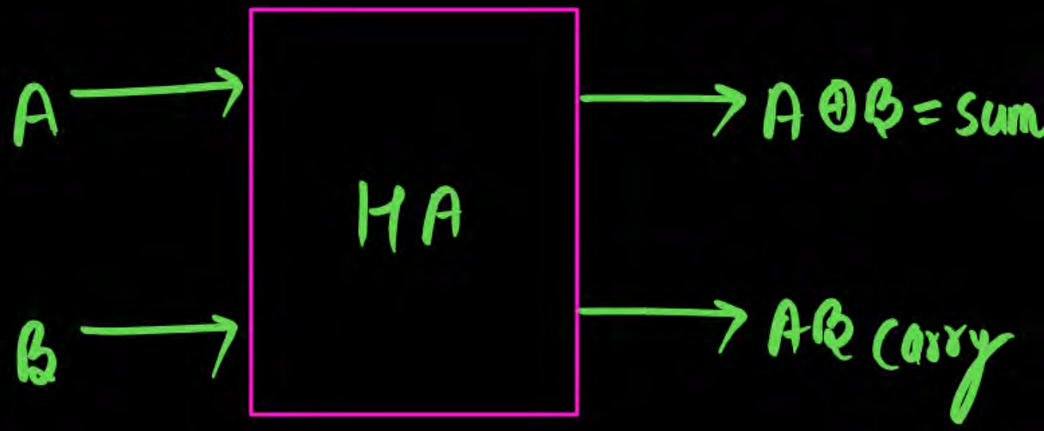
$$\text{sum} = \overline{A}B + A\overline{B} = A \oplus B$$
$$\text{carry} = AB$$

Step 4. Minimization

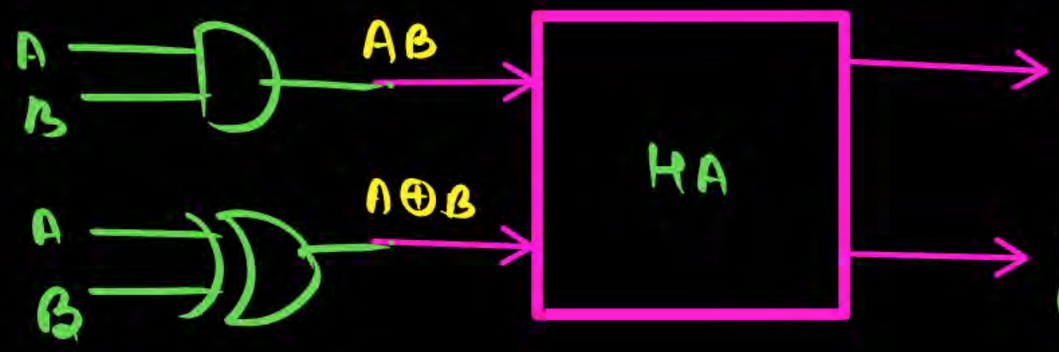
Half Adder

Step 5.





Q Find y_0 & y_1 = ?



$y_0 = AB \oplus (A \oplus B) = \overline{AB} \cdot (A \oplus B) + AB \cdot \overline{A \oplus B}$

$= (\bar{A} + \bar{B})[\bar{A}B + A\bar{B}] + AB[\bar{A}\bar{B} + AB]$

$\Rightarrow \bar{A}B + A\bar{B} + AB$

$\Rightarrow \bar{A}B + A[\bar{B} + B]$

$\Rightarrow \bar{A}B + A = (\bar{A} + A)(A + B)$

$= A + B$ Ans

$y_1 = AB \cdot (A \oplus B)$

$= AB[\bar{A}B + A\bar{B}]$

$= AB \cdot \bar{A}B + AB \cdot A\bar{B}$

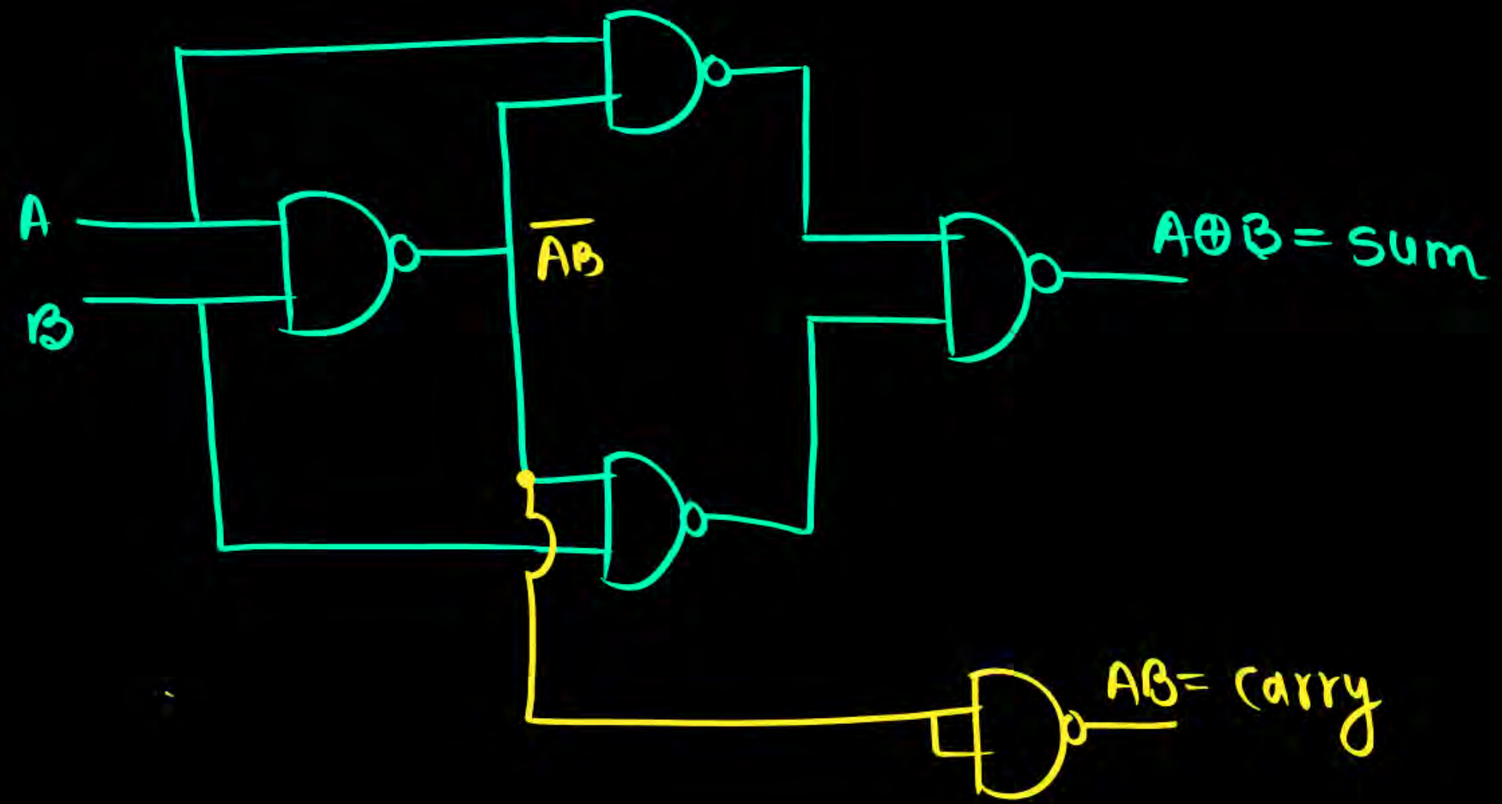
$= 0 + 0$

$= 0$

ZQ.

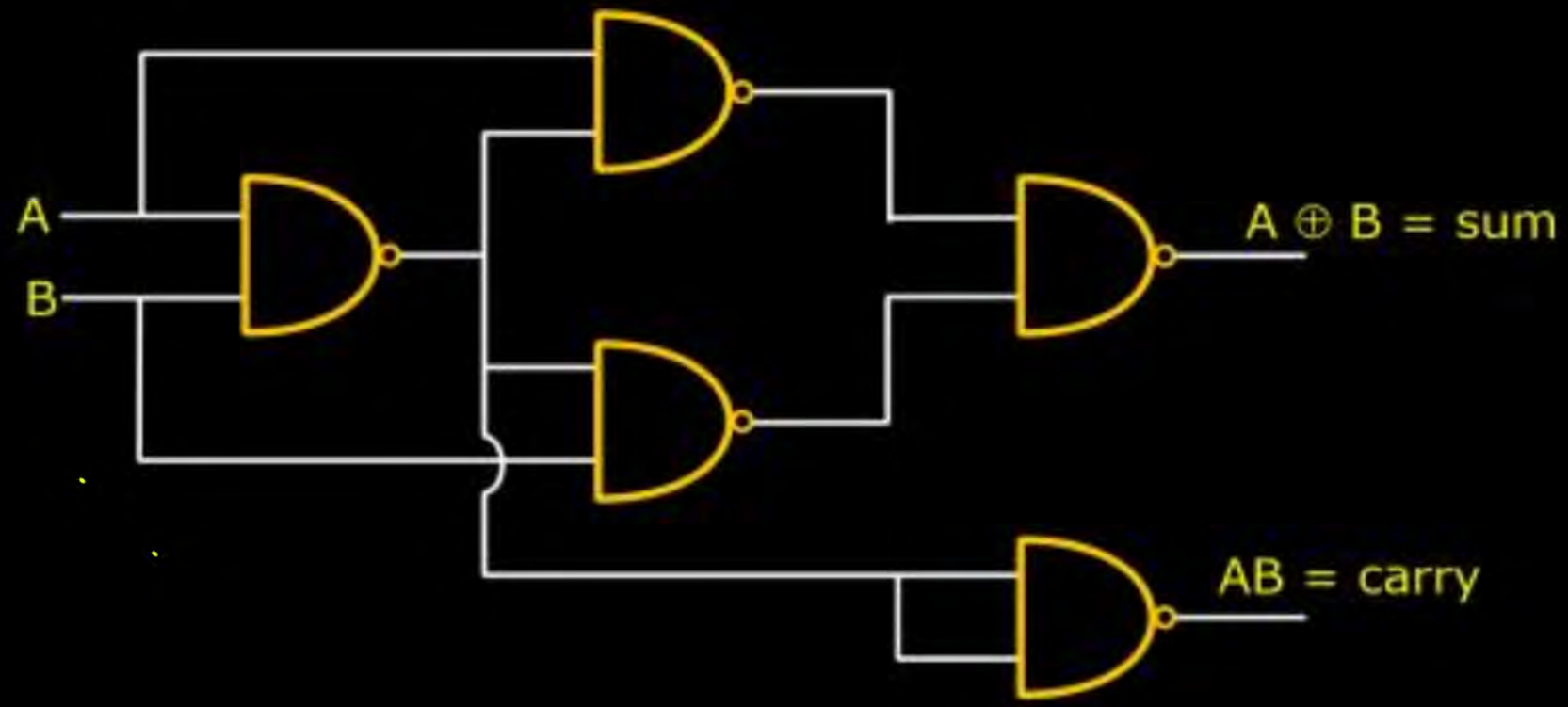
NOTE $AB \oplus (A \oplus B) = A + B$

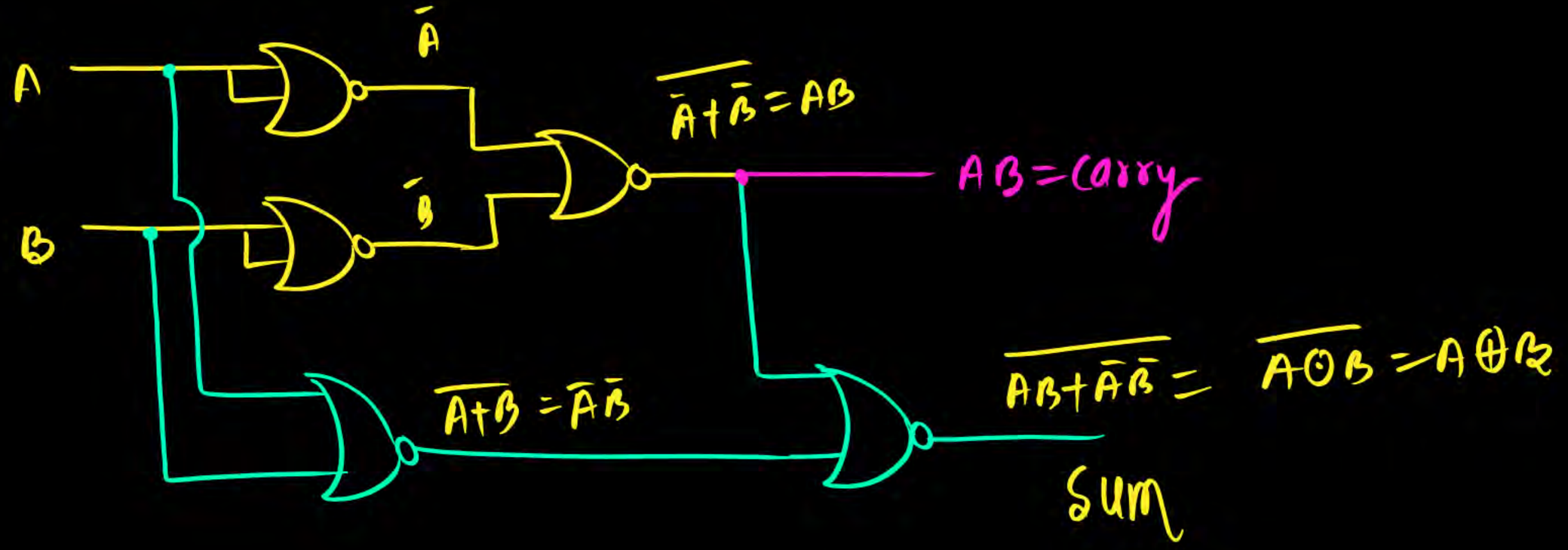
$ABC \oplus (A \oplus B \oplus C) = A + B + C$


A
B
$\overline{AB}$
$A \oplus B$ = Sum
AB = Carry

Half Adder

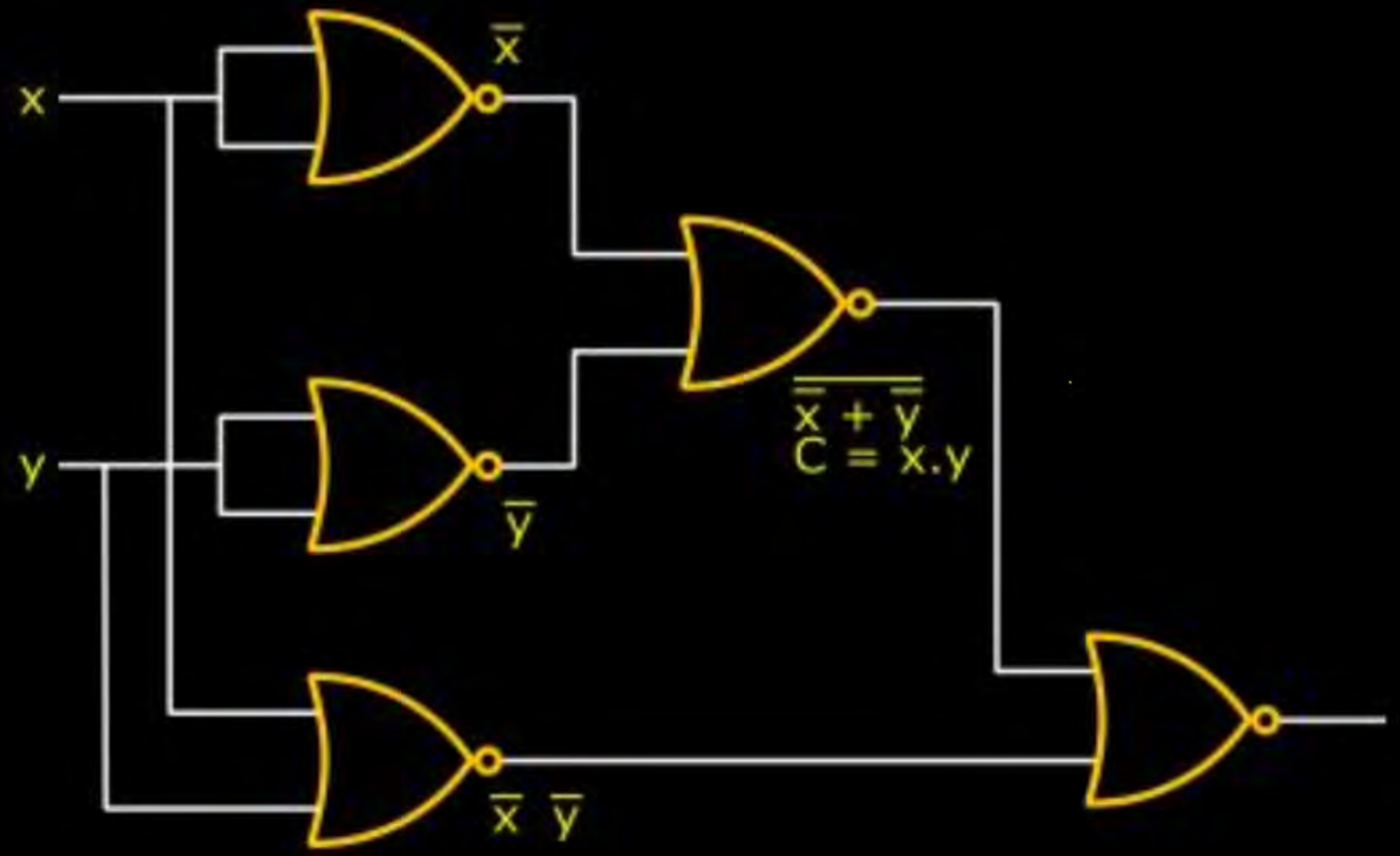
By NAND GATE




A
B
$\bar{A}$
$\bar{B}$
$\overline{\bar{A}+\bar{B}} = AB$
AB = Carry
$\overline{A+B} = \bar{A}\bar{B}$
$\overline{AB+\bar{A}\bar{B}} = \overline{A \odot B} = A \oplus B$
Sum

Half Adder

By NAND GATE



HALF

Adder

Subtractor

NAND/NOR = 5

FULL

Adder, Subtractor

NAND/NOR = 9

FULL ADDER :→

$$\begin{array}{r} 0 \\ 0 \\ +\ 0 \\ \hline 0\ 0 \end{array} \qquad \begin{array}{r} 0 \\ 0 \\ +\ 1 \\ \hline 0\ 1 \end{array} \qquad \begin{array}{r} 0 \\ 1 \\ +\ 1 \\ \hline 1\ 0 \end{array} \qquad \begin{array}{r} 1 \\ 1 \\ +\ 1 \\ \hline 1\ 1 \end{array}$$

Full Adder

Three bit adder are known as full adder

Step 1.



Full Adder

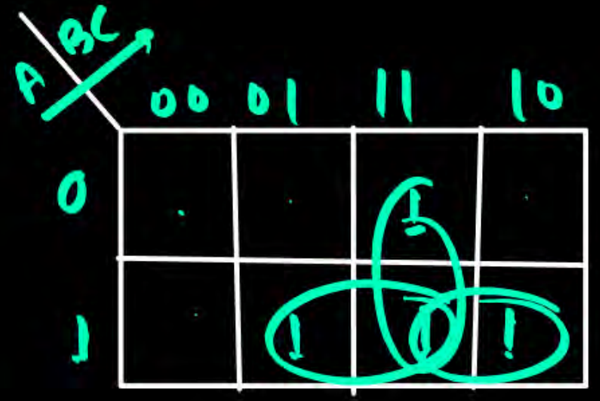
Step 2.

	A	B	C	Sum	Carry
0 →	0	0	0	0	0
1 →	0	0	1	1	0
2 →	0	1	0	1	0
3 →	0	1	1	0	1
4 →	1	0	0	1	0
5 →	1	0	1	0	1
6 →	1	1	0	0	1
7 →	1	1	1	1	1

Majority Input high Logic

Full Adder

Step 3. Sum (A, B, C) = $\Sigma m(1,2,4,7) = A \oplus B \oplus C$



Step 4 :

$$\text{Carry}(A,B,C) = \Sigma m(3,5,6,7) \quad \text{—— (1)}$$

$$= \bar{A}BC + A\bar{B}C + AB\bar{C} + ABC$$

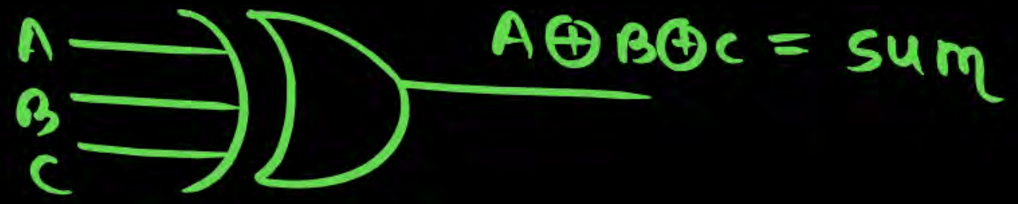
$$= (\bar{A}B + A\bar{B})C + AB(\bar{C} + C)$$

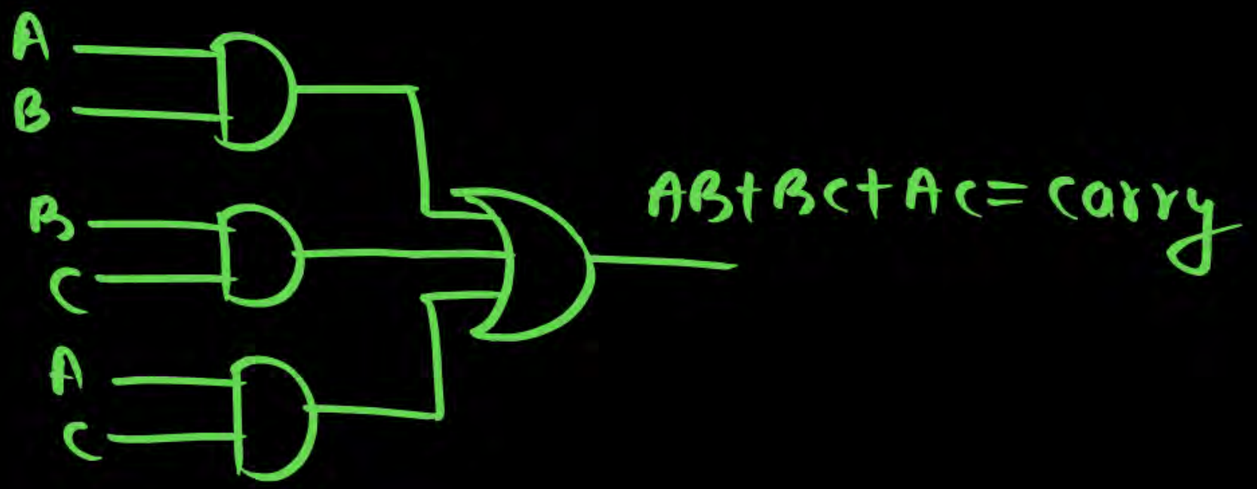
$$= (A \oplus B)C + AB \rightarrow \text{Semiminimized} \quad \text{—— (2)}$$

$$= AB + BC + AC \quad \text{—— (3)}$$

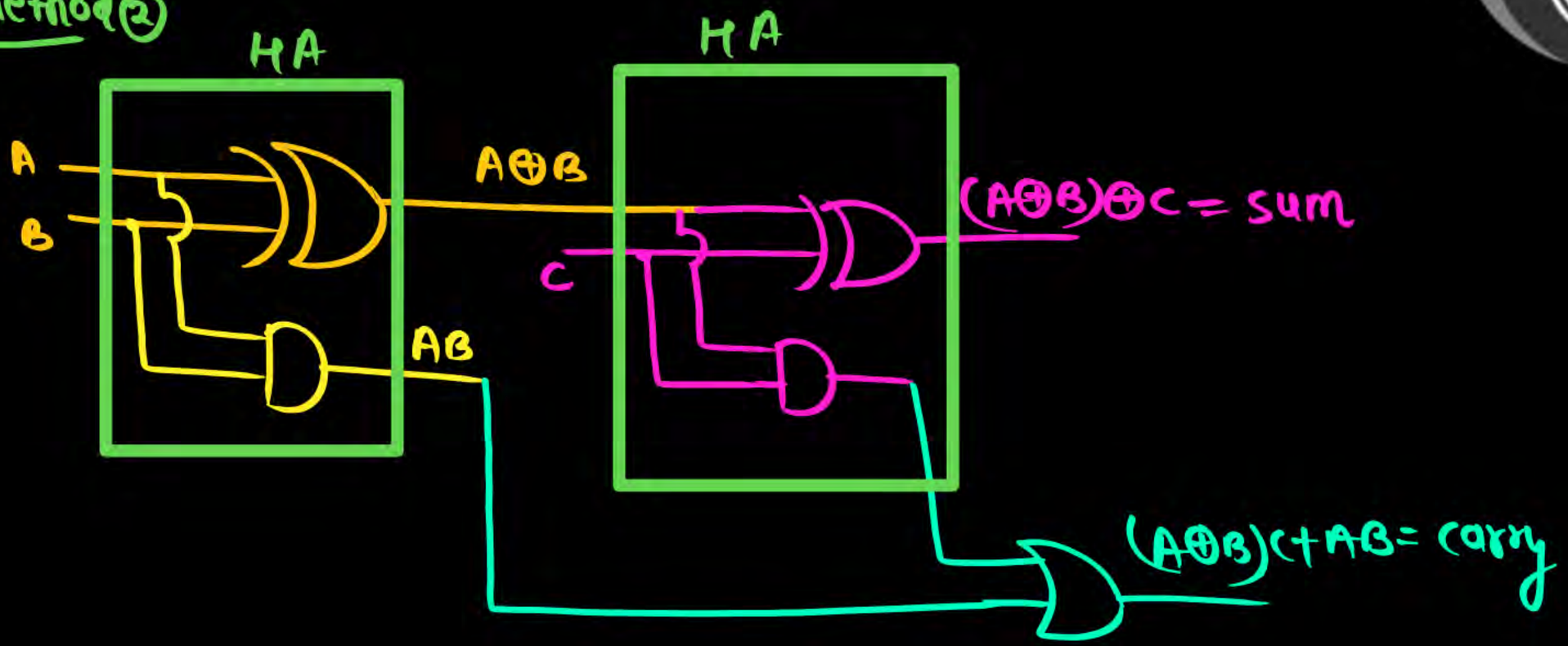
Step 5:

Method 1.



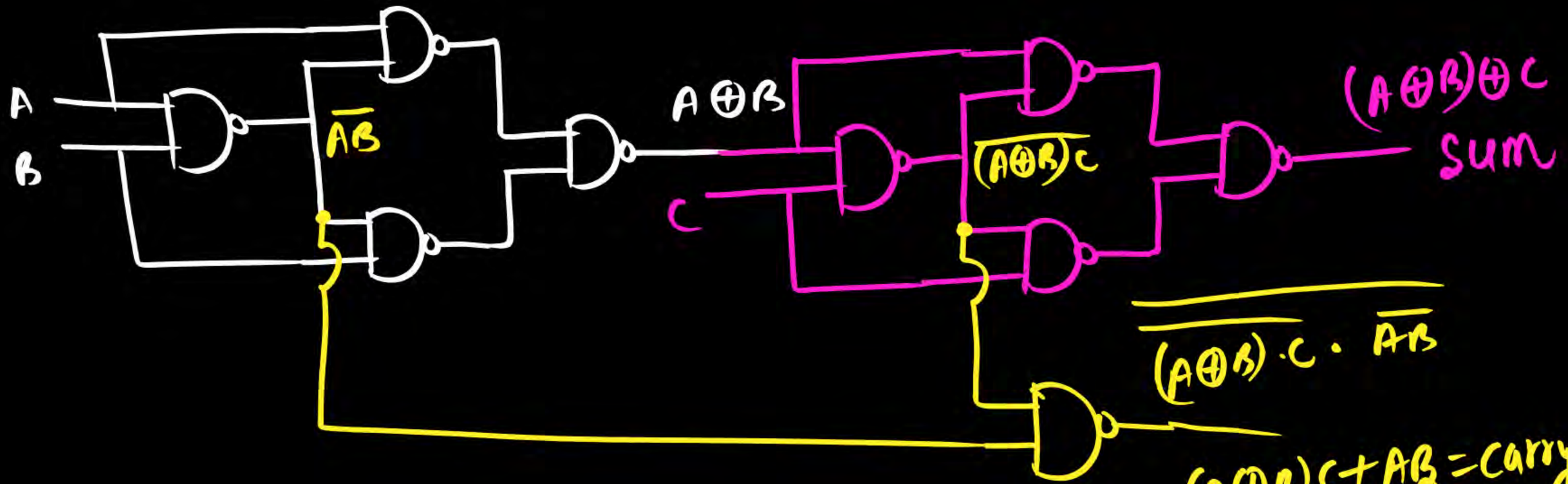


Method ②



1. FULL ADDER = 2 HALF ADDER + 1 OR GATE

FULL ADDER BY USING NAND :→

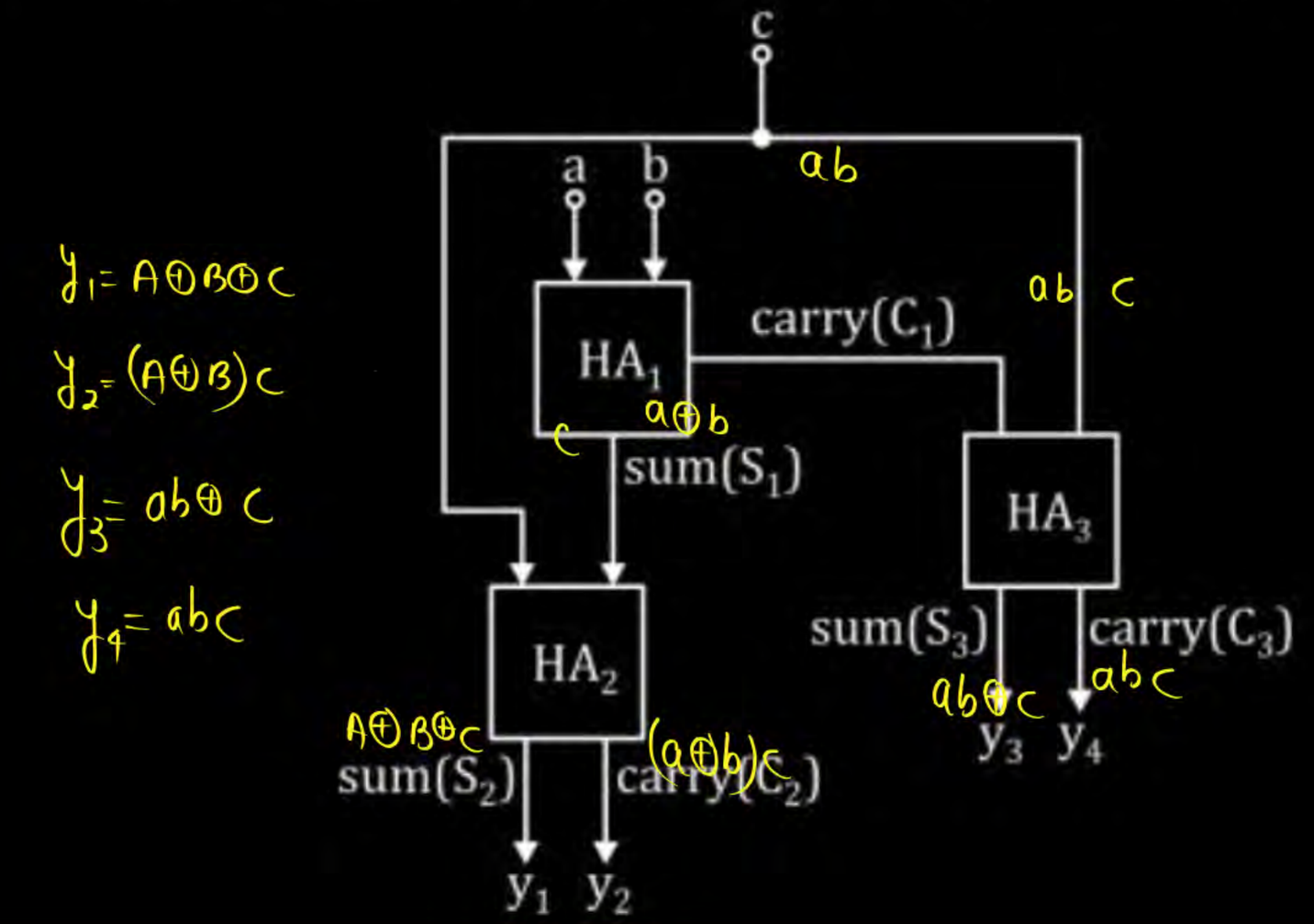


By using NOR :-

?

Question **(MCQ)**

#Q. Three half adders HA_1, HA_2 and HA_3 are inter-coupled as shown below. The four output functions y_1, y_2, y_3 and y_4 are expressed in terms of inputs a, b and c. Which one of the following output expressions is correct?



A $y_1 = (a \oplus b)\, c$

B $y_2 = (a \oplus b) \oplus c$

C $y_3 = ab \oplus c$

D $y_4 = a\,(b \oplus c)$

$y_1 = A \oplus B \oplus C$

$y_2 = (A \oplus B)C$

$y_3 = ab \oplus c$

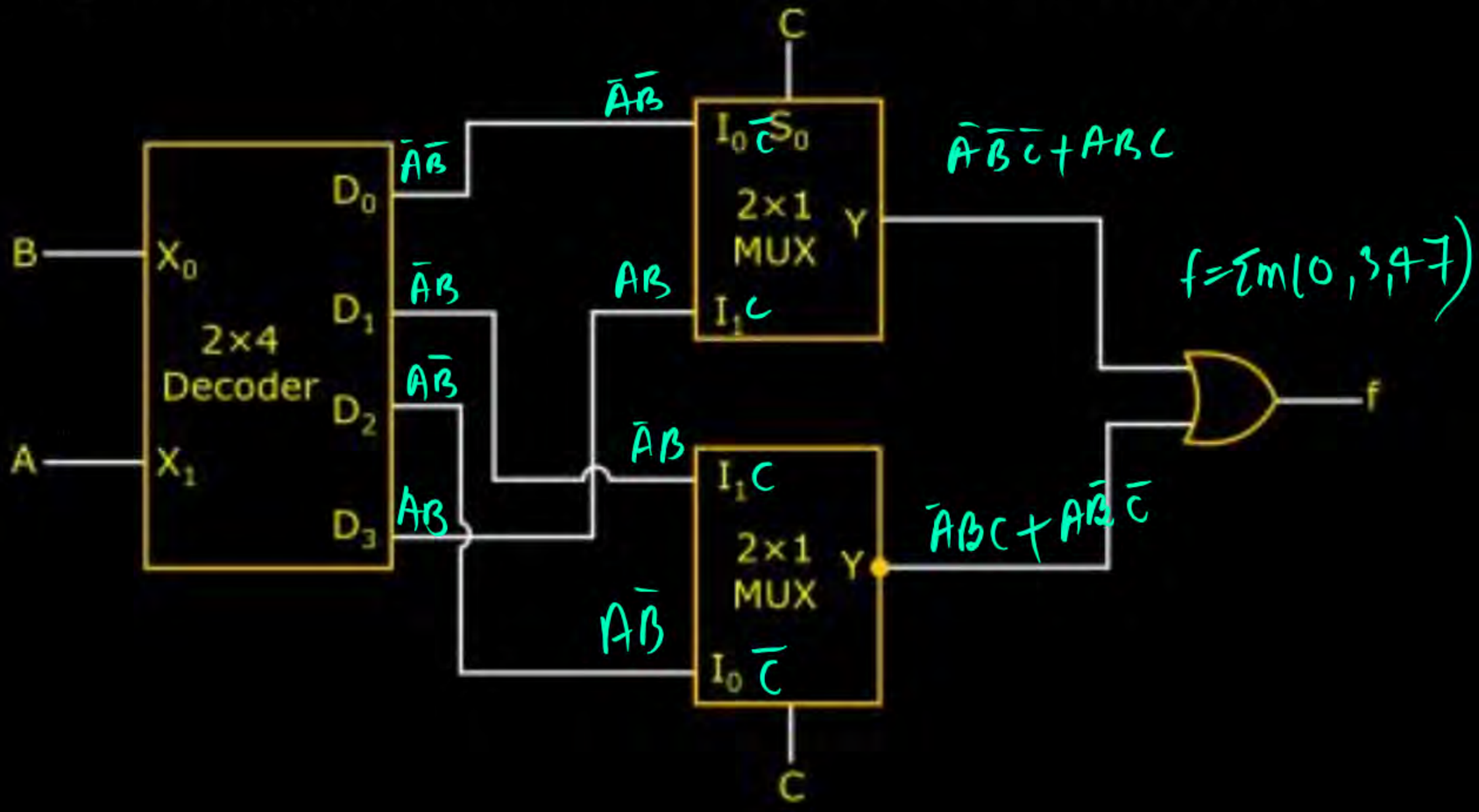
$y_4 = abc$

Question **(MCQ)**

A full adder is implemented with two half adders and one OR gate. OR gate is used to derive the final carry function of full adder. In each half adder, T_{sum} = 25 ns and T_{carry} = 20 ns and T_{OR} = 25 ns. The minimum time required to derive both the sum and carry function of a full adder after applying the inputs is _____ ns

Question **(MCQ)**

A logic function 'f' is implemented by the circuit shown in the figure below. The circuit consists of one 2×4 decoder, two 2×1 multiplexers and a two input or gate connected in cascade. Then the function f is equal to.



A. $A \oplus B$

B. $A \oplus B \oplus C$

C. $B \odot C$

D. $A \odot B$

$\bar{A}\bar{B}\bar{C} + A\bar{B}\bar{C} + \bar{A}BC + ABC$

$\bar{B}\bar{C}(\bar{A}+A) + BC(\bar{A}+A)$

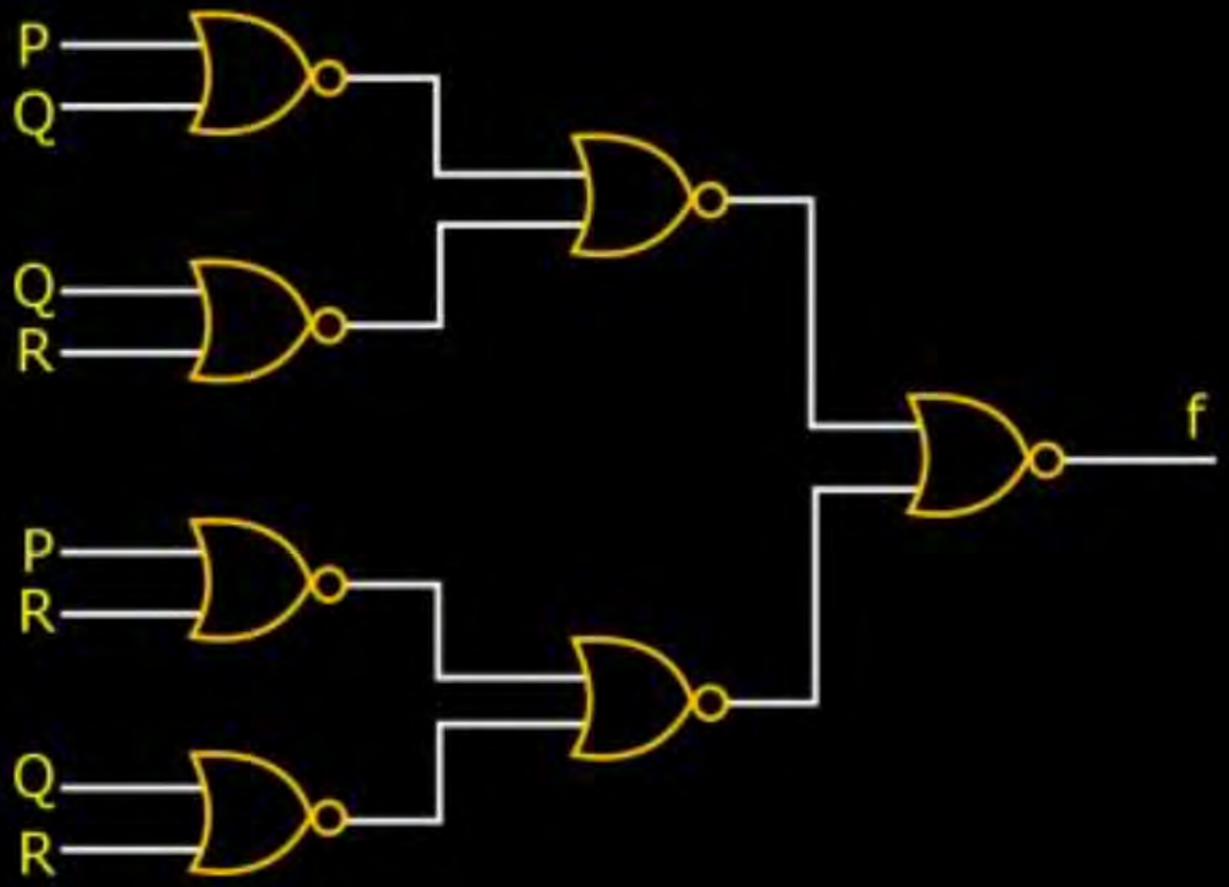
$\bar{B}\bar{C} + BC$

$B \odot C$

A

Question **(MCQ)**

What is the Boolean expression for the output f of the combinational logic circuit of NOR gates given below?



A $\overline{Q+R}$

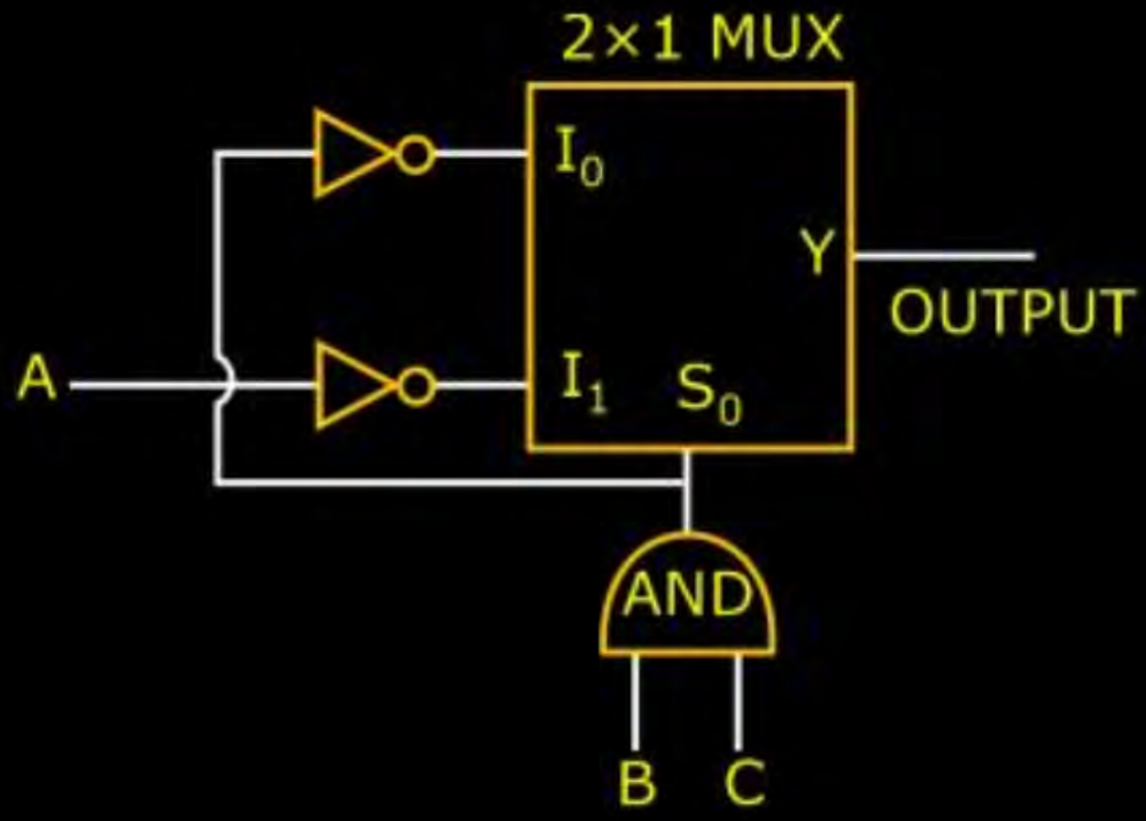
B $\overline{P+Q}$

C $\overline{P+R}$

D $\overline{P+Q+R}$

Question **(MCQ)**

The combinational circuit given below implements which of the following



A NOR gate

B XOR gate

C NAND gate

D None of these

Question **(MCQ)**

A digital circuit is as given below :

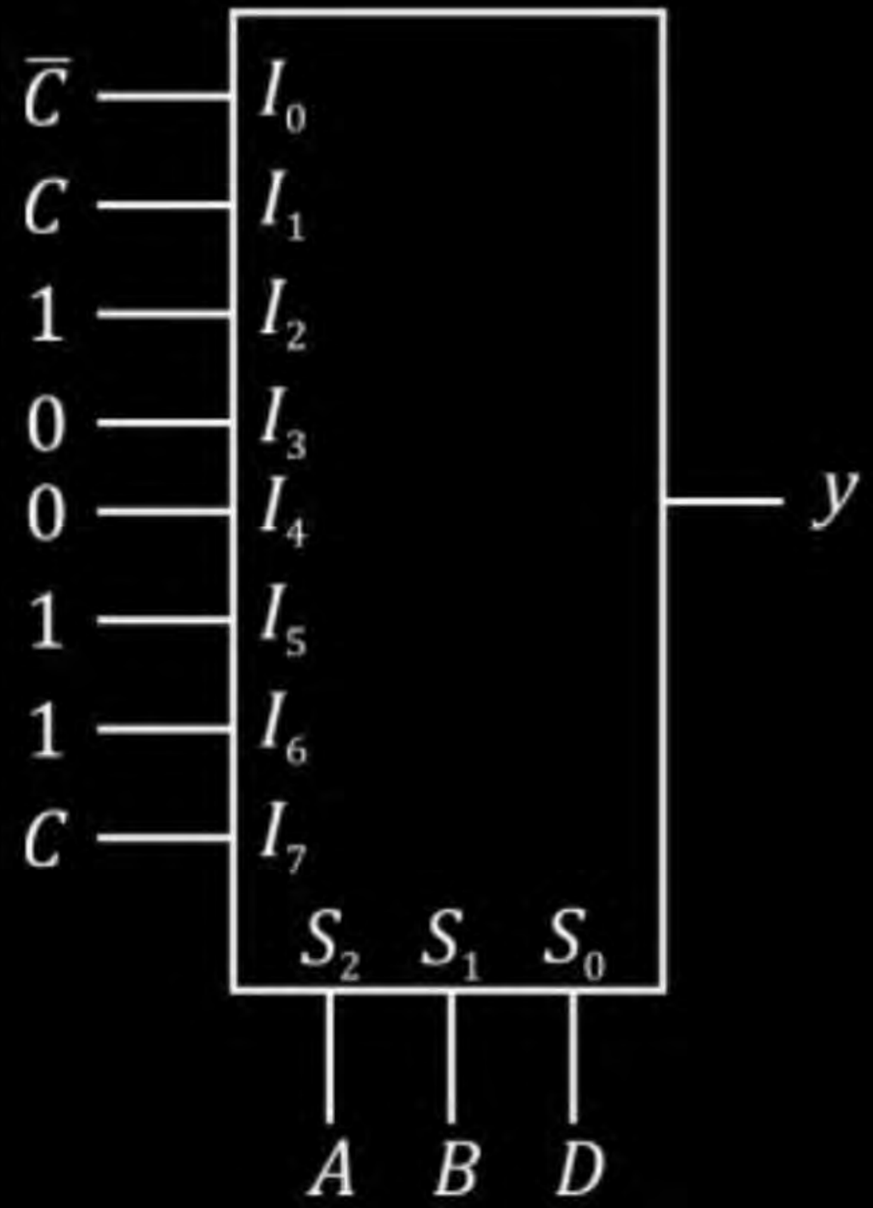
Output y will be :

(a) $y = \Sigma\,(0,3,4,6,7,9,11,12,14,15)$

(b) $y = \Sigma\,(0,1,3,4,6,8,9,11,12,14,15)$

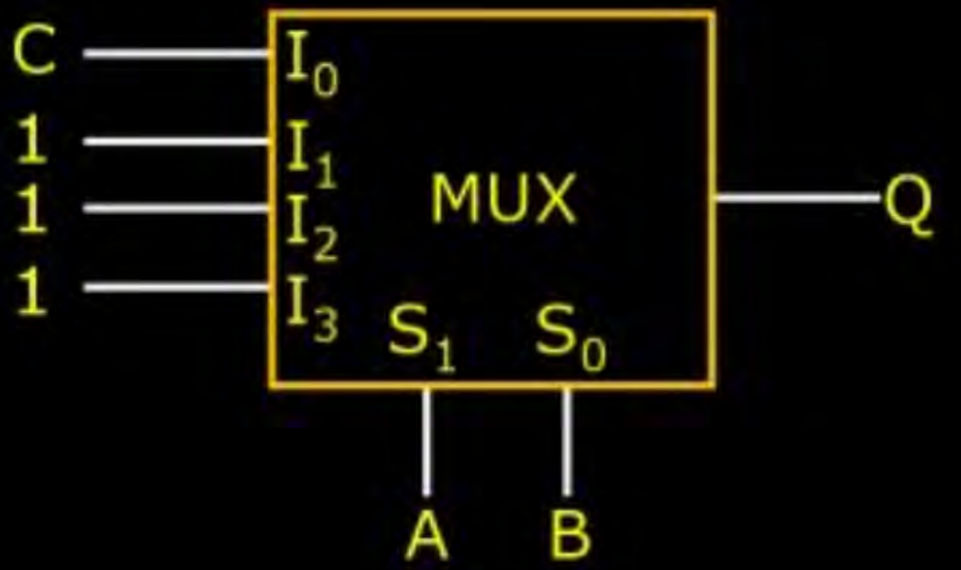
(c) $y = \Sigma\,(0,3,4,6,9,11,12,14,15)$

(d) $y = \Sigma\,(0,3,4,6,9,10,11,12,14,15)$



Question **(MCQ)**

The combinational logic circuit shown in the given figure has an output Q which is



A. ABC

B. A + B + C

C. $A \oplus B \oplus C$

D. A.B + C

Question **(MCQ)**

A logic circuit consist of two 2 × 4 decoders as shown in the figure. The output of decoder are as follow:

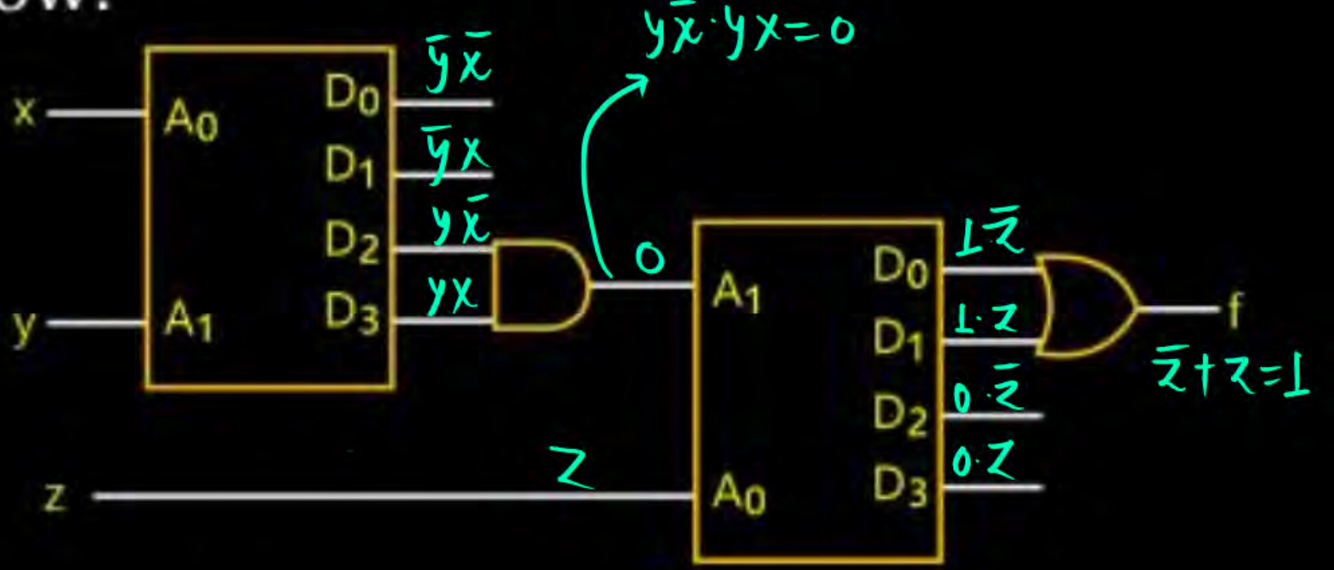
D0 = 1 when A0 = 0, A1 = 0

D1 = 1 when A0 = 1, A1 = 0

D2 = 1 when A0 = 0, A1 = 1

D3 = 1 when A0 = 1, A1 = 1

The value of f(x, y, z) is



A_1	A_0	D_3	D_2	D_1	D_0
0	0	0	0	0	1
0	1	0	0	1	0
1	0	0	1	0	0
1	1	1	0	0	0

A 0

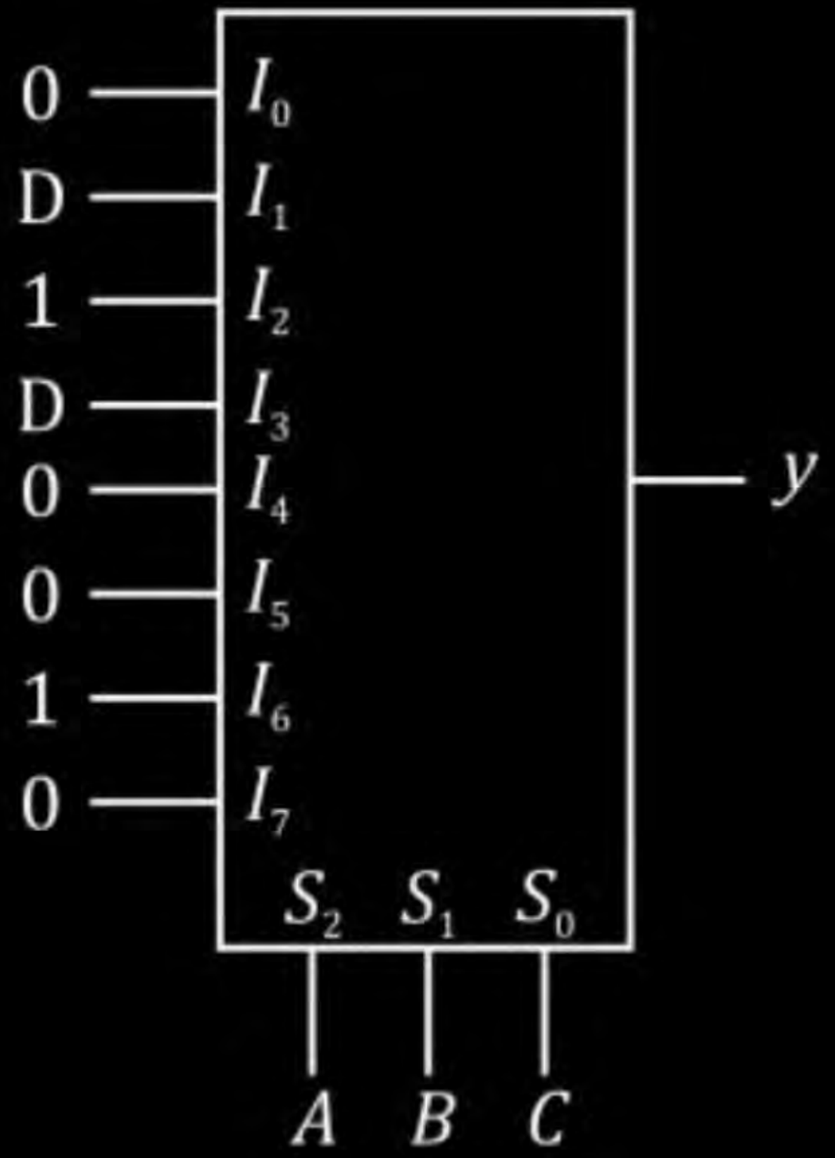
B z

C $\bar{z}$

D 1

Question **(MCQ)**

A digital circuit is as given below :



$y\,(A, B, C, D)$ will be

A $A\bar{B}C + A\bar{C}D$

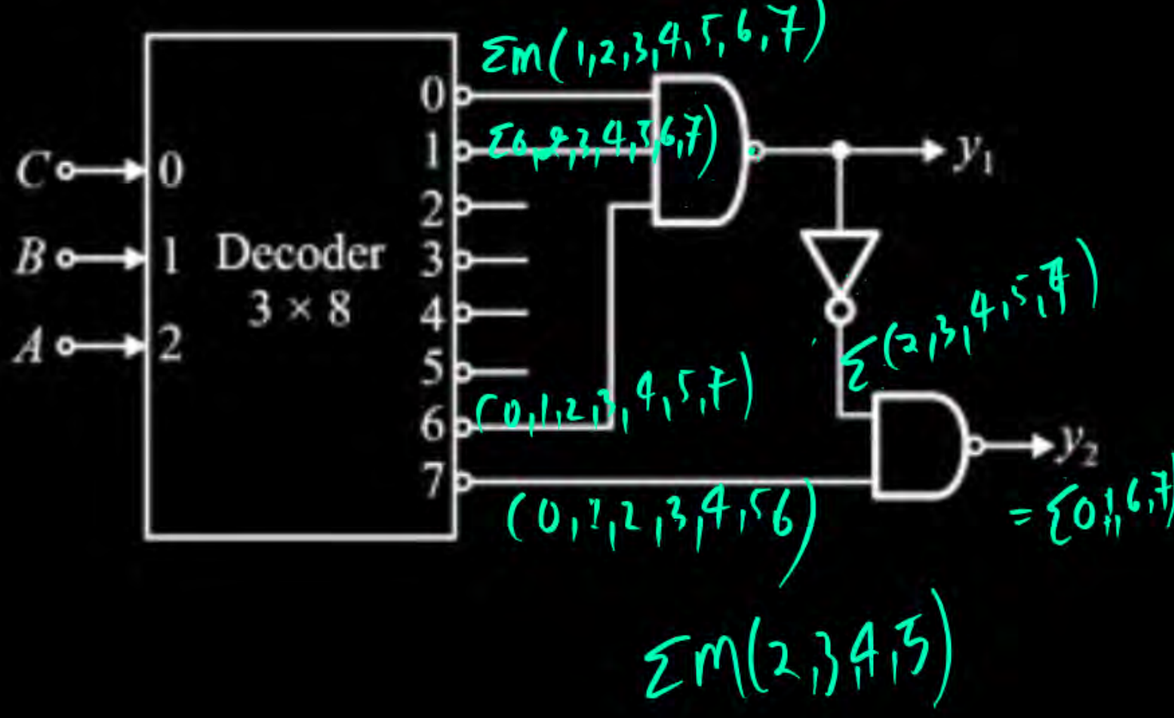
B $BC + A\bar{B}D$

C $B\bar{C} + \bar{A}CD$

D $\bar{B}C + A\bar{B}D$

Question [MCQ]

A 3 line to 8 line decoder with three inputs A, B, C and two outputs y_1 and y_2, is configured as shown below. The minimized expression of outputs will be

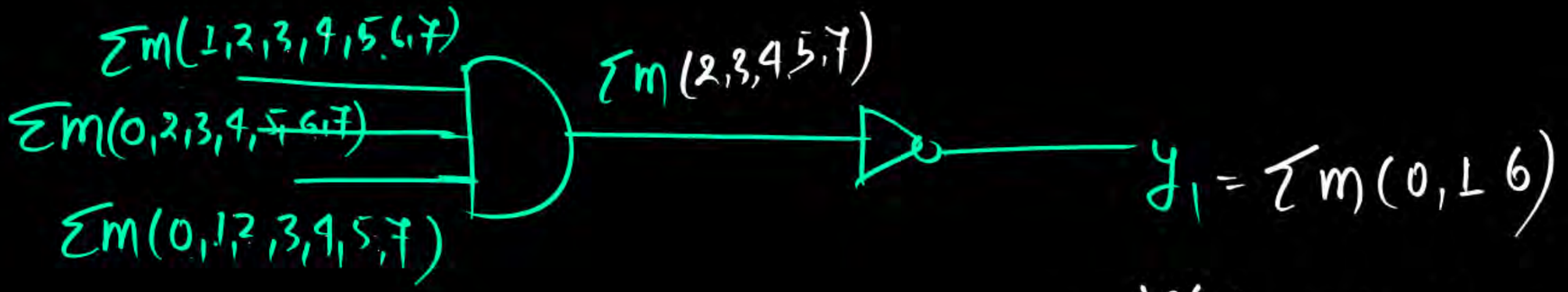


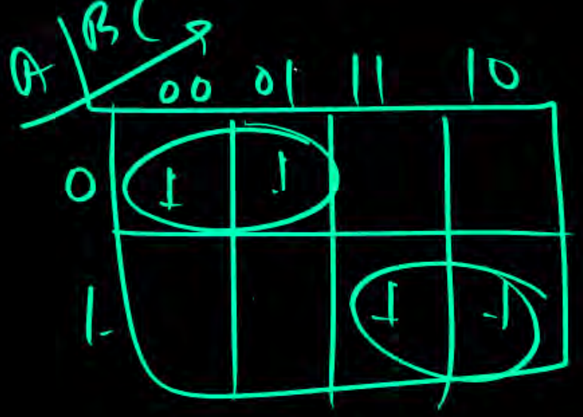
A $y_1 = \bar{A}\bar{B} + AB\bar{C};\ y_2 = \overline{A \oplus B}$

B $y_1 = AB + \bar{A}\bar{B}C;\ y_2 = A \oplus B$

C $y_1 = \bar{A}B + A\bar{C};\ y_2 = AB + AC$

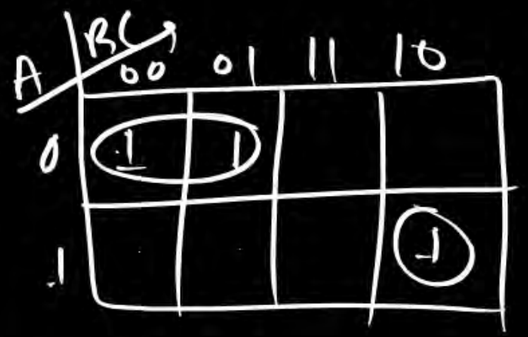
D $y_1 = A\bar{B} + \bar{A}C;\ y_2 = \bar{A}B + \bar{B}C$





$\bar{A}\bar{B} + AB$

$A \odot B$

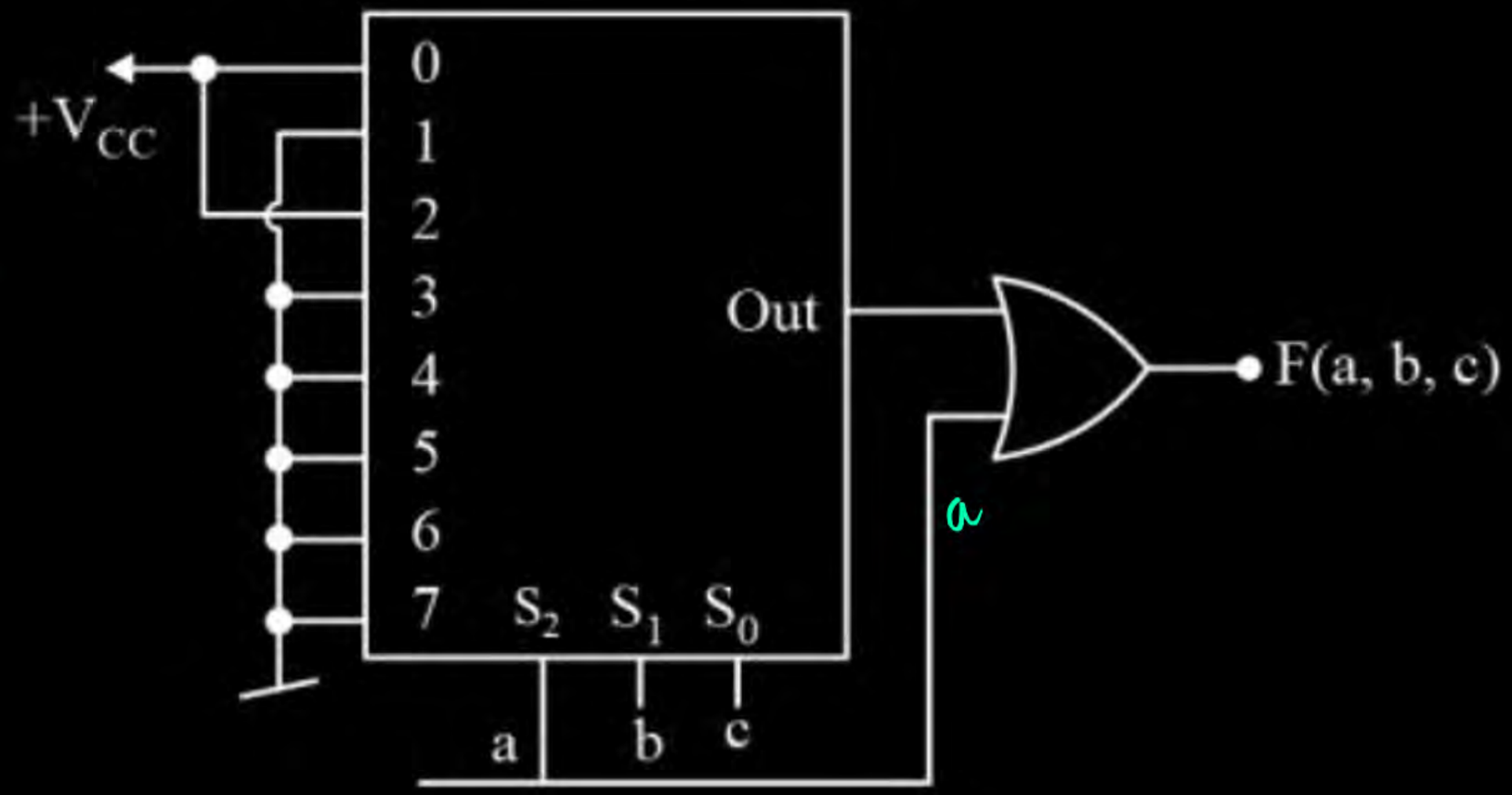


$\bar{A}\bar{B} + AB\bar{C}$

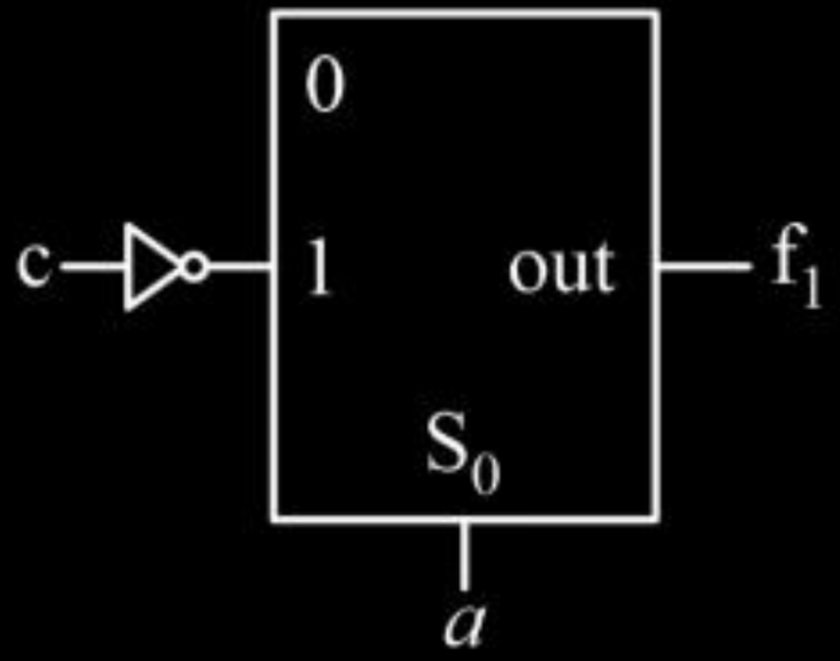
Question [MCQ]

★★☆

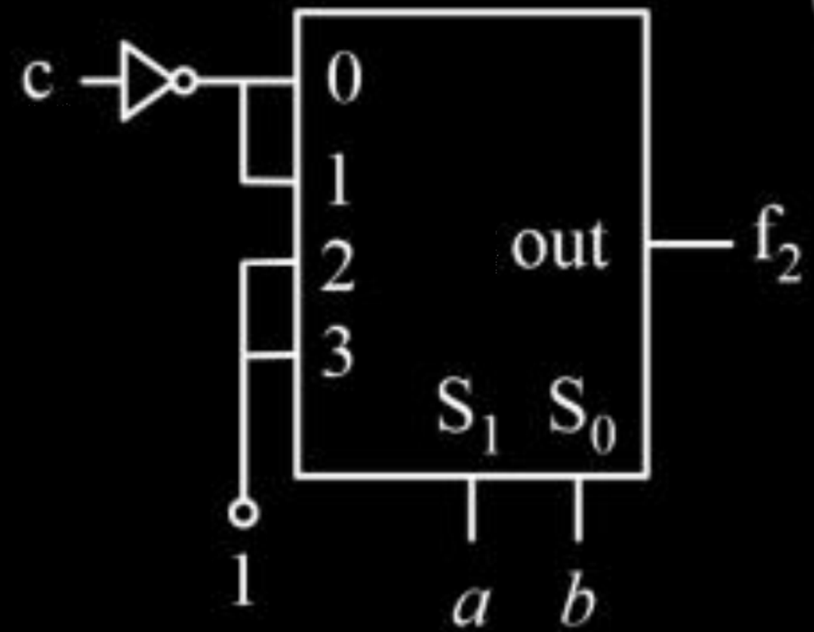
An 8 ×1 [MUX is configured as shown below to implement a function f(a, b, c), a is MSB. Alternatively, the same function can also be implemented using smaller size 2 × 1 MUX and 4 × 1 MUX. The correct alternative is



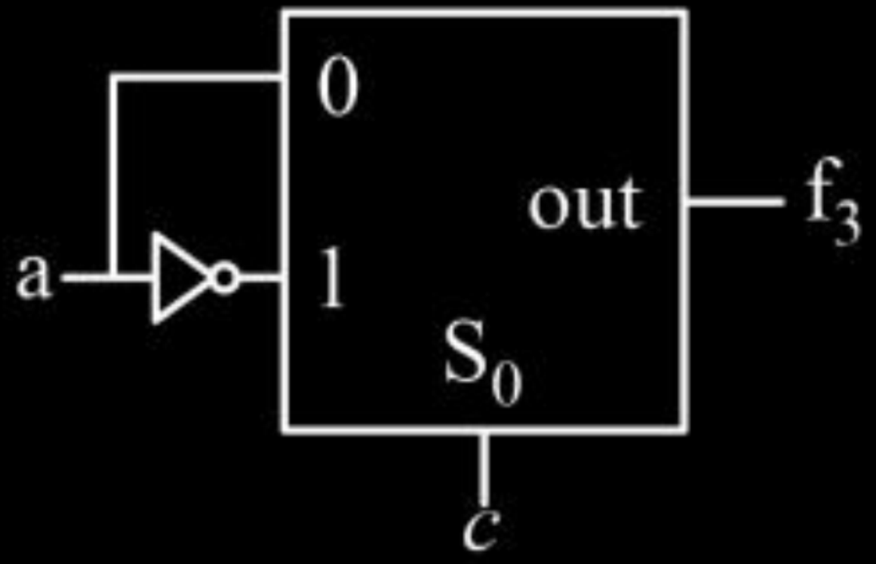
A


0
1
out
f_1
c
S_0
a

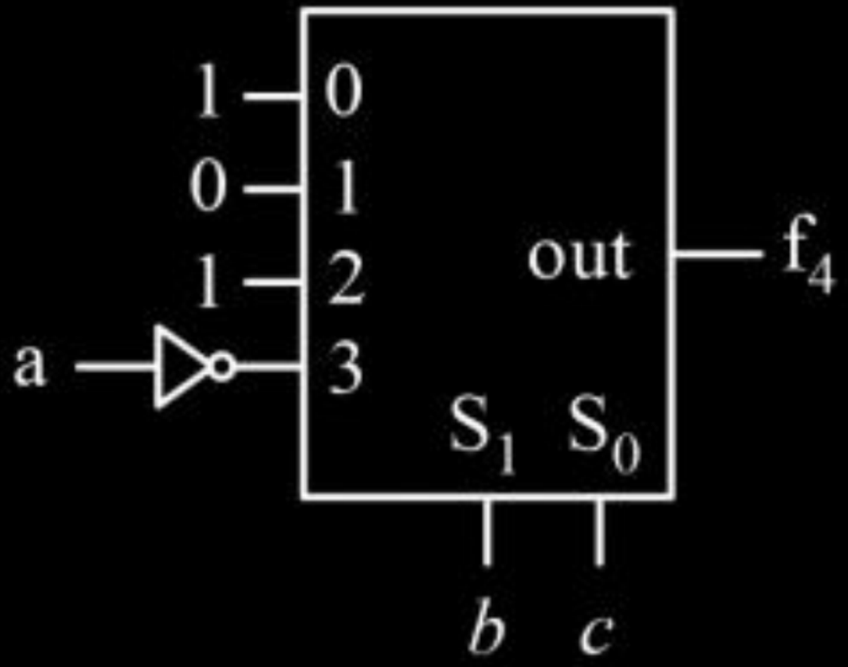
B


c
0
1
2
3
out
f_2
S_1 S_0
1
a b

C


0
out
f_3
a
1
S_0
c

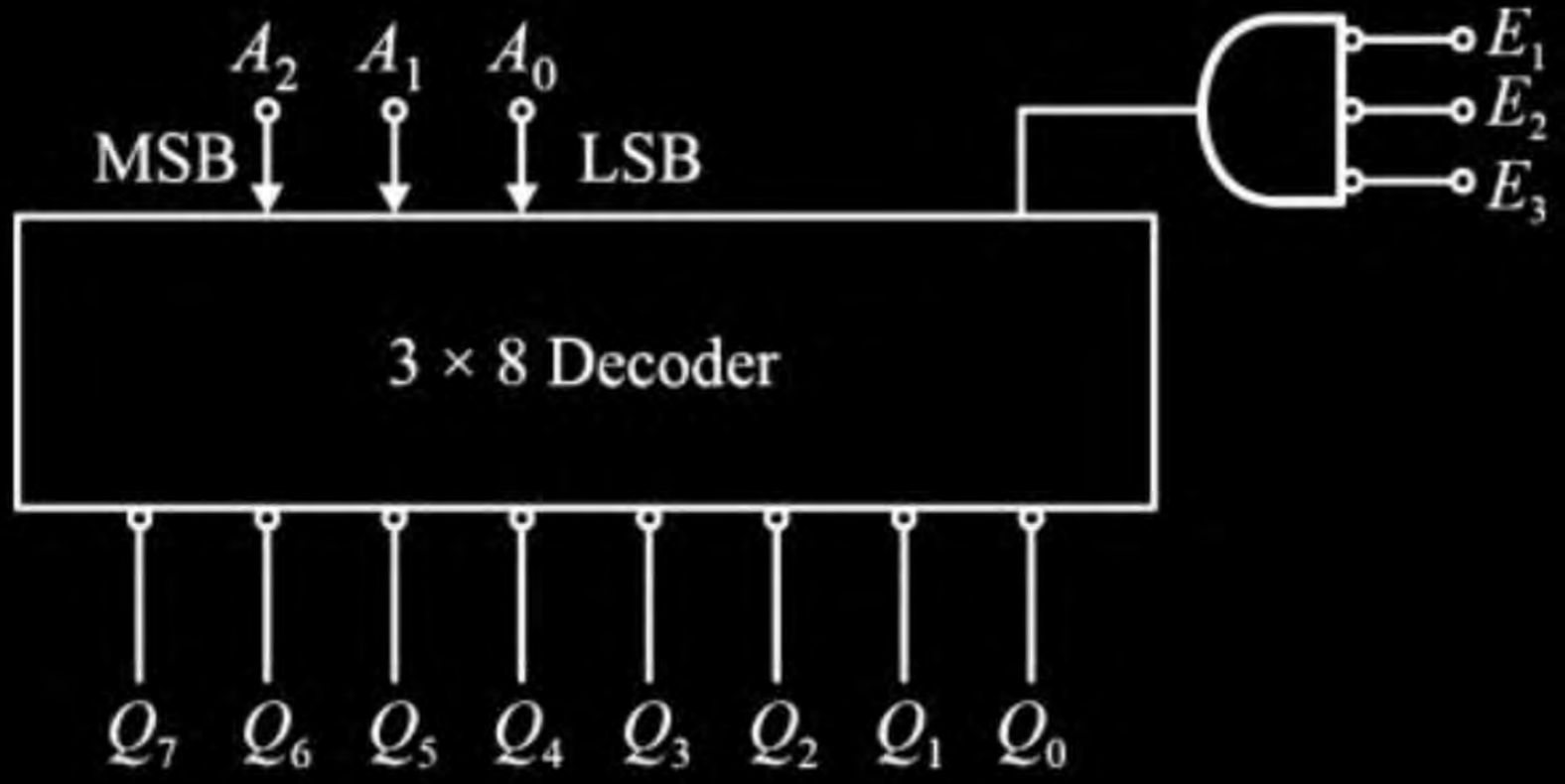
D


1
0
1
0
1
2
3
out
f_4
a
S_1 S_0
b c

Question [MCQ]

The logic diagram of a 3 × 8 decoder with active low outputs is shown below.

What is state of outputs Q_7, ... Q_0 for the set of inputs $E_3 = E_1 = 1$, $E_2 = 0$, $A_2 = A_1 = 1$ and $A_0 = 0$?



A 1111 1111

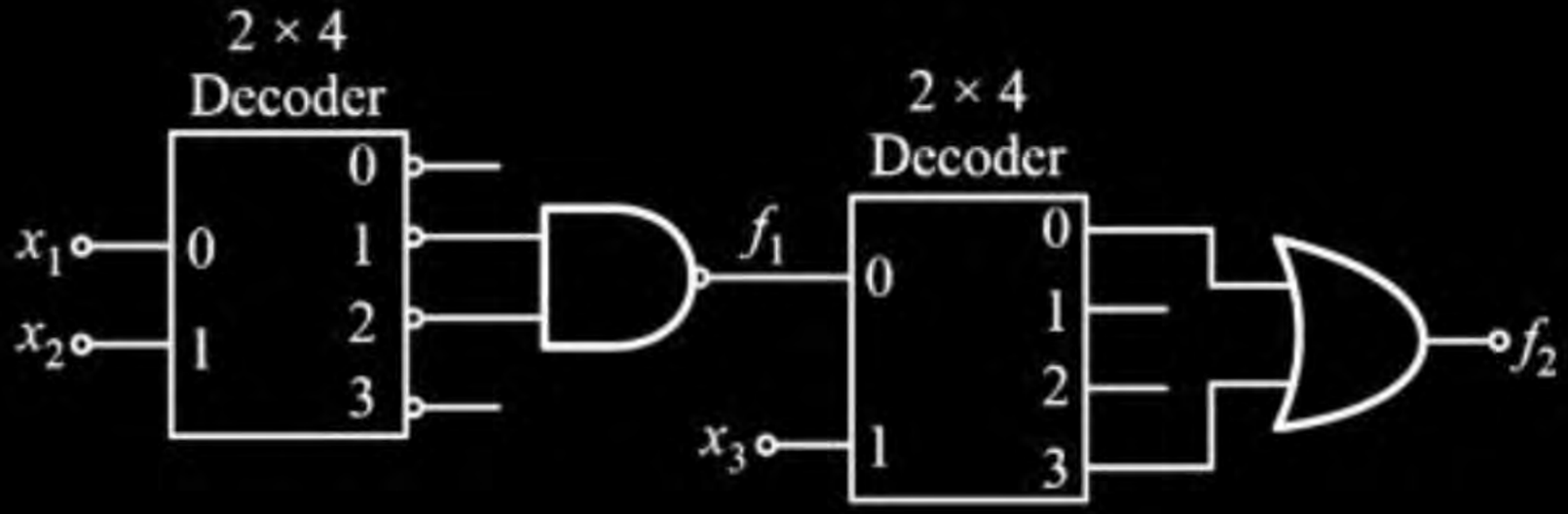
B 1011 1111

C 1111 0111

D 0000 0000

Question [MCQ]

Two 2 × 4 decoders one with active low outputs and another with active high output are interconnected as shown below. The output function $f_2(x_3, x_2, x_1)$ will be



A $f_2 = (x_1 \oplus x_2) \odot x_3$

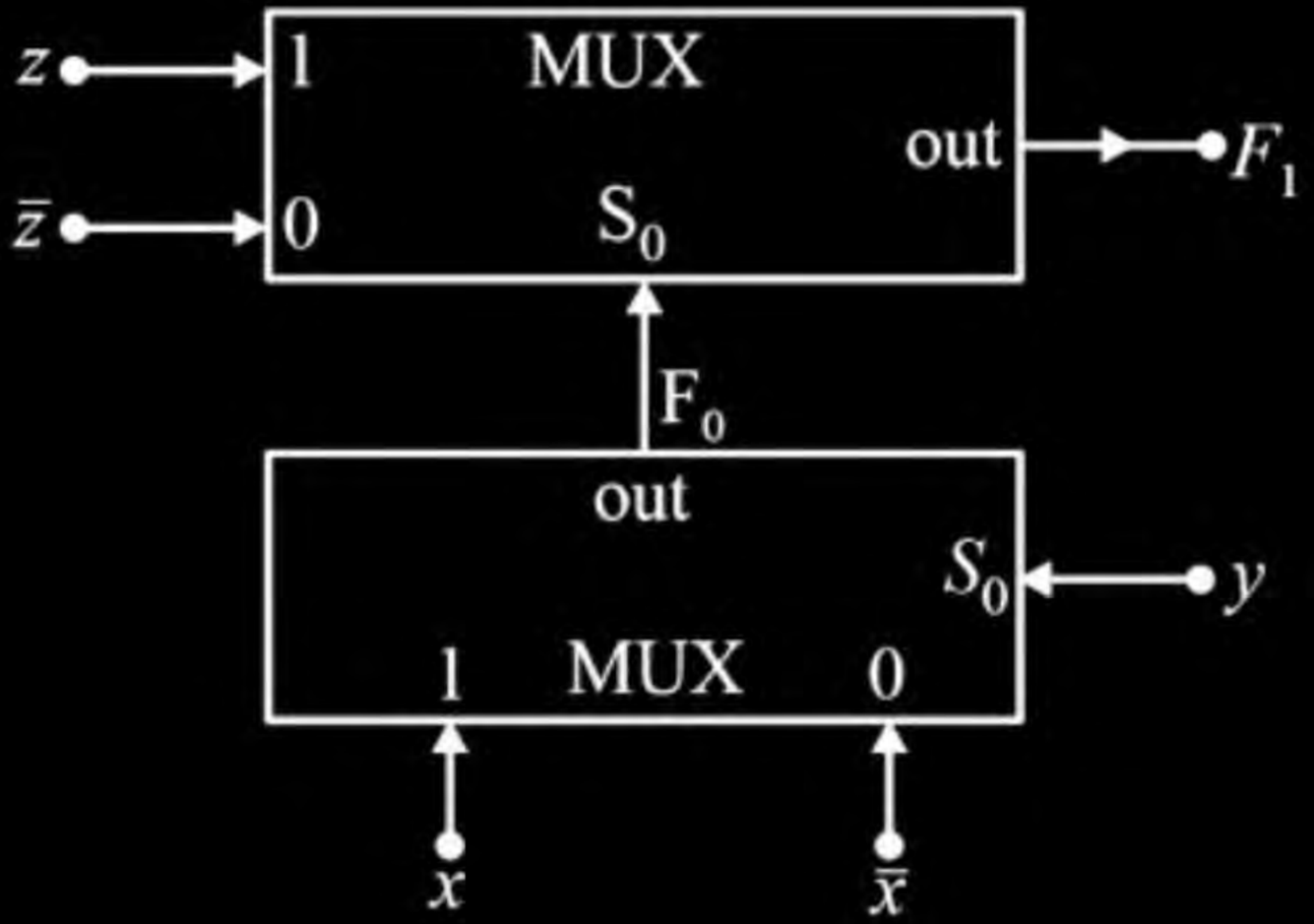
B $f_2 = (x_1 \odot x_2) \odot x_3$

C $f_2 = (x_1 \oplus x_2) \oplus x_3$

D $f_2 = (x_1 \oplus x_2) \oplus x_3$

Question [MCQ]

In the multiplexer configuration shown below, the output F_1 will be



A $\overline{(x \oplus y)} \oplus z$

B $\overline{\overline{(x \oplus y)} + z}$

C $(x \oplus y) \oplus \bar{z}$

D $(x \oplus y) \oplus z$

Thank you DOSTO